FOR A MORPHISM OF GROUPOIDS OVER A FIELD AS IN THE MORPHISM-GROUPOIDS SITUATION, AN OPEN IMAGE IS CLOSED

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Allowed standard prerequisites (editorial).

- The morphism of groupoids over a field situation fixing R_1, R_2, a, s_i, t_i, c_i (more-groupoids chapter, Situation situation-morphism-groupoids-on-field, canonical tag 04Q5; kind reference). cited at statement lines 1711-1712. Supplies the notation and assumptions in which the statement is phrased.
- Base change to a field extension making source and target isomorphisms at an adapted point (more-groupoids chapter, Lemma lemma-groupoid-on-field-explain-points, canonical tag 04LR; collected on this lane as L22505). cited at proof lines 1718-1720. Provides the adapted pair k inside k' and r_2' over r_2 used to reduce to the case where s_2 and t_2 induce isomorphisms on residue fields.
- Restriction of a groupoid over a field to a field extension has surjective open structure maps on the arrow scheme side (more-groupoids chapter, Lemma lemma-restrict-groupoid-on-field, canonical tag 04LQ). cited at proof lines 1732-1733. Makes p_2 : R'_2 to R_2 surjective and open, so closure is detected after base change.
- Open surjective morphisms and closure: preimages of closures are closures of preimages (topology chapter, Lemma topology-lemma-open-morphism-quotient-topology, canonical tag 02YB). cited at proof lines 1734-1735. Transfers r_2 in the closure of a(R_1) to r_2' in the closure of a'(R'_1) along the open surjective p_2.
- Automorphism moving the identity to a point when source and target are isomorphisms (more-groupoids chapter, Lemma lemma-groupoid-on-field-move-point, canonical tag 04LS; collected on this lane as L22506). cited at proof lines 1741-1743. Shows c(r_2, a(R_1)) is an open neighbourhood of r_2, opening the way for the closure-intersection and composition argument.

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Source reference. Original reference 04Q5: sources/stacks-chow/tags.txt, source lines 6138–6138. Cited at statement lines 1711-1712: notation and assumptions fixing R_1, R_2, the morphism a and the structure maps.

Source reference. Original lemma 04LR: sources/stacks-chow/tags.txt, source lines 6017–6017. Cited at proof lines 1718-1720: supplies the adapted extension k inside k' and the point r_2' over r_2 for the base-change reduction.

Source reference. Original lemma 04LQ: sources/stacks-chow/tags.txt, source lines 6016–6016. Cited at proof lines 1732-1733: p_2 : R'_2 to R_2 is surjective and open.

Source reference. Original lemma 02YB: sources/stacks-chow/tags.txt, source lines 3870–3870. Cited at proof lines 1734-1735: r_2 in the closure of a(R_1) implies r_2' in the closure of a'(R'_1) along the open surjective p_2.

Source reference. Original lemma 04LS: sources/stacks-chow/tags.txt, source lines 6018–6018. Cited at proof lines 1741-1743: c(r_2, a(R_1)) is an open neighbourhood of r_2 in the residue-field-isomorphism case.

Lemma 0.2. *Notation and assumptions as in Situation 04Q5. If $a(R_1)$ is open in R_2, then $a(R_1)$ is closed in R_2.*

Proof. Let $r_2 \in R_2$ be a point in the closure of $a(R_1)$. We want to show $r_2 \in a(R_1)$. Pick $k \subset k'$ and $r_2' \in R_2'$ adapted to (U, R_2, s_2, t_2, c_2) and r_2 as in Lemma 04LR. Let R_i' be the restriction of R_i via the morphism $U' = \mathrm{Spec}(k') \to U = \mathrm{Spec}(k)$. Let $a' : R_1' \to R_2'$ be the base change of a. The diagram

$$\begin{array}{ccc} R_1' & \xrightarrow{a'} & R_2' \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle p_2} \\ R_1 & \xrightarrow{a} & R_2 \end{array}$$

is a fibre square. Hence the image of a' is the inverse image of the image of a via the morphism $p_2 : R_2' \to R_2$. By Lemma 04LQ the map p_2 is surjective and open. Hence by Topology, Lemma 02YB we see that r_2' is in the closure of $a'(R_1')$. This means that we may assume that $r_2 \in R_2$ has the property that the maps $k \to \kappa(r_2)$ induced by s_2 and t_2 are isomorphisms.

In this case we can use Lemma 04LS. This lemma implies $c(r_2, a(R_1))$ is an open neighbourhood of r_2. Hence $a(R_1) \cap c(r_2, a(R_1)) \neq \emptyset$ as we assumed that r_2 was a point of the closure of $a(R_1)$. Using the inverse of R_2 and R_1 we see this means $c_2(a(R_1), a(R_1))$ contains r_2. As $c_2(a(R_1), a(R_1)) \subset a(c_1(R_1, R_1)) = a(R_1)$ we conclude $r_2 \in a(R_1)$ as desired. $\square$